

① ① $t = 10s$

$$V = V_0 e^{-\frac{t}{RC}} \cdot 10$$

$$= 32 e^{-\frac{10}{27 \times 10^3 \times 220 \times 10^{-6}}}$$

$$= 5.94V$$

$$I = \frac{V}{R} = \frac{5.94}{27 \times 10^3} = 2.2 \times 10^{-4} A$$

$$Q = CV$$

$$= 220 \times 10^{-6} \times 5.94$$

$$= 1.31 \times 10^{-3} C$$

② ② $t = 10s$

$$U = \frac{1}{2} CV^2$$

$$= \frac{1}{2} \times 220 \times 10^{-6} \times (5.94)^2$$

$$= 3.89 \times 10^{-3} J$$

③

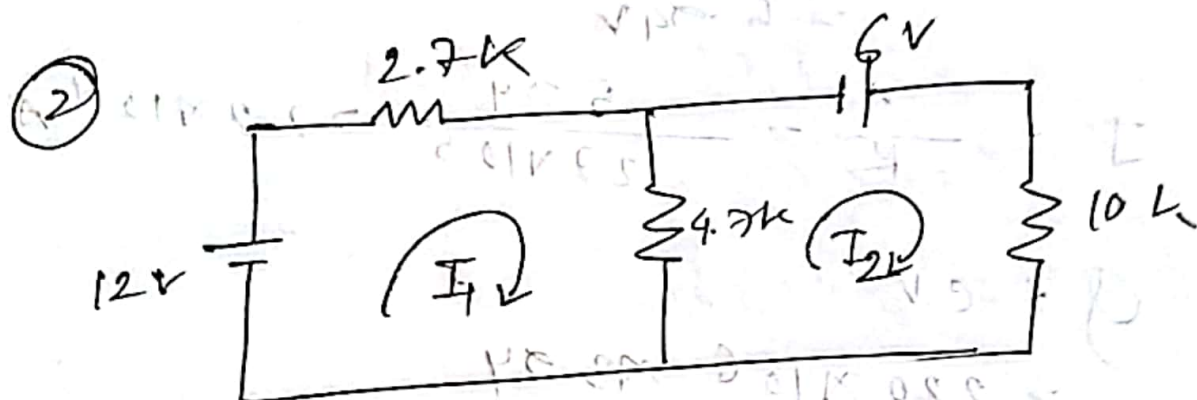
$$V_C = V_0 e^{-t/RC}$$

$$\Rightarrow \frac{V_0}{2} = V_0 e^{-t/RC}$$

$$\Rightarrow \frac{1}{2} = e^{-t/RC}$$

$$\Rightarrow I = 0.67 \times 27 \times 10^3 \times 220 \times 10^{-6}$$

$$= 4.115 \text{ A}$$



$$-12 + 2.7I_1 + 4.7(I_1 - I_2) = 0$$

$$\Rightarrow 7.4I_1 - 4.7I_2 = 12 \quad \text{--- (1)}$$

$$\Rightarrow -6 + 10I_2 + 4.7(I_2 - I_1) = 0$$

$$\Rightarrow 14.7I_2 - 4.7I_1 = 6 \quad \text{--- (2)}$$

① and ② $\Rightarrow$

$$I_1 = 2.36 \text{ mA}$$

$$I_2 = 1.16 \text{ mA}$$

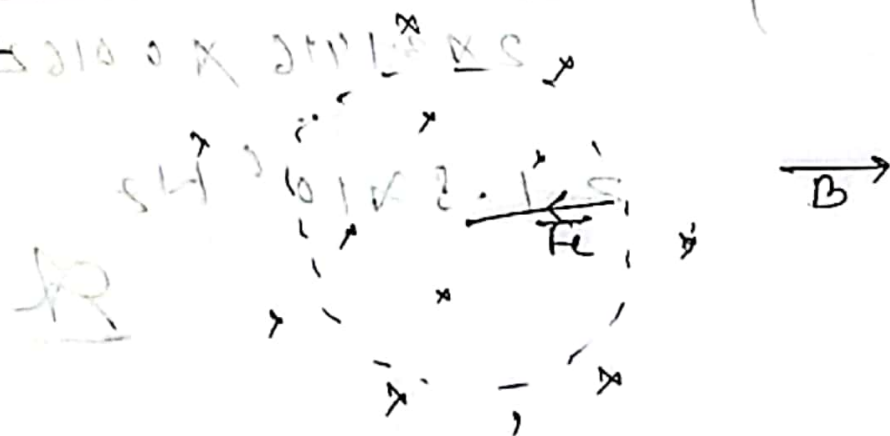
through

$$R_1 \therefore I_1 = 2.36 \text{ mA}$$

$$R_2 \therefore I_2 = 1.2 \text{ mA}$$

$$R_3 \therefore I_3 = 1.16 \text{ mA}$$

③ A₂



④ A

$$F_B = qvB \sin \theta$$

$$\Rightarrow v = \frac{3.6 \times 10^{-15}}{1.6 \times 10^{-19} \times 0.12}$$

$$= 1.87 \times 10^5 \text{ m/s}$$

⑤ $F = \frac{mv^2}{r}$

$$\Rightarrow r = \frac{mv^2}{F} = \frac{1.67 \times 10^{-27} \times (1.87 \times 10^5)^2}{3.6 \times 10^{-15}}$$

$$= 0.0162 \text{ m}$$

again,

$$\frac{1}{T} = \frac{v}{2\pi R}$$

$$\Rightarrow f = \frac{1.88 \times 10^5}{2 \times 3.1416 \times 0.0162}$$

$$= 1.8 \times 10^6 \text{ Hz}$$

A